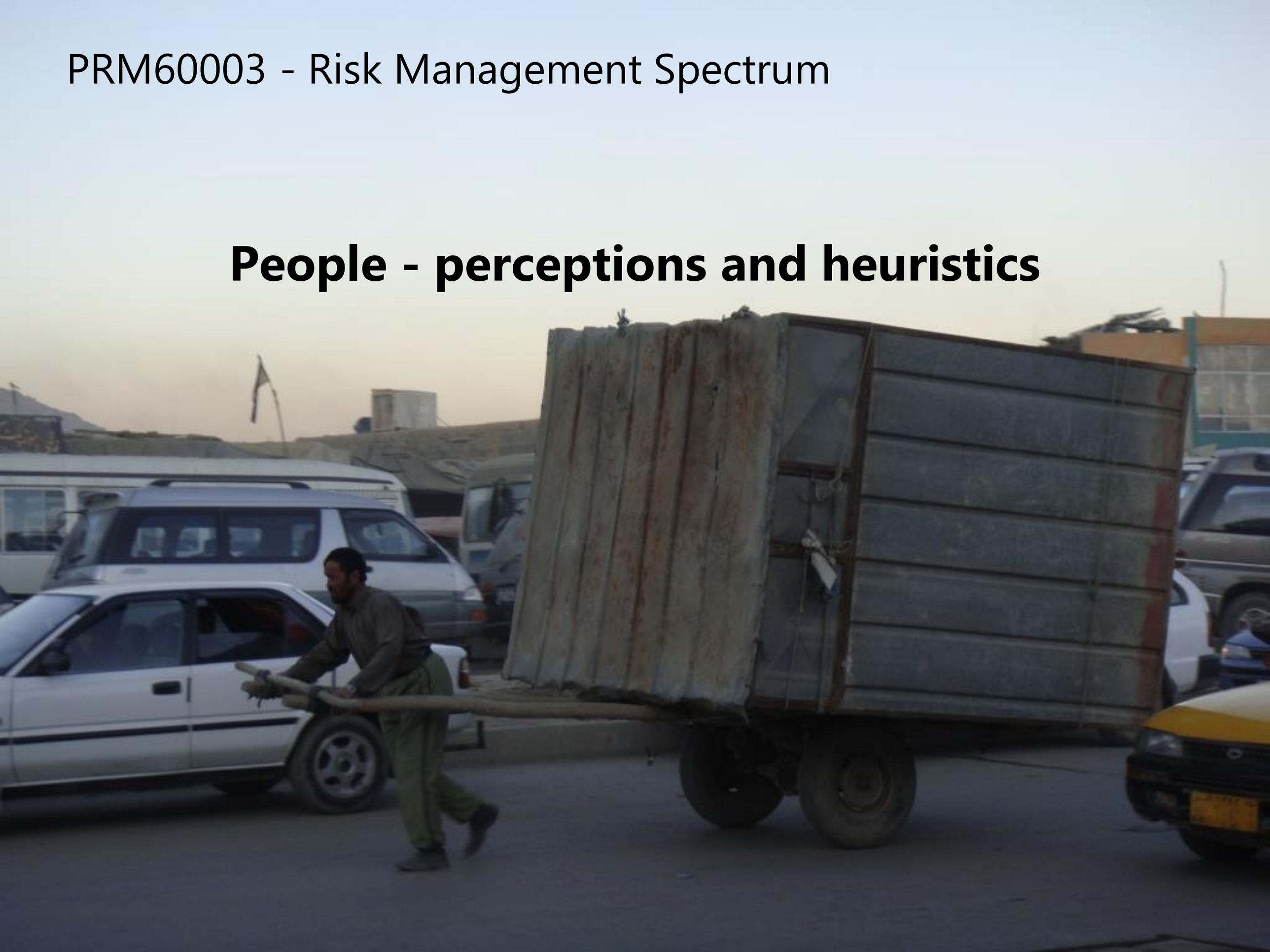
PRM60003 - Risk Management Spectrum

People - perceptions and heuristics



People

Given the standardisation of risk management, we can be lured into viewing risk management as a mechanistic process. Once the identification, evaluation and measurement of risks are tabulated, project managers can feel confident in their decisions and predictions.

However, how good are we really at forecasting risk?
Ultimately, risks are managed by humans, not objects. A flashing light on the dashboard is only an indication of a threat or hazard.

Our perception of risk can have a serious impact on risk management.
The following table compares descriptive and critical ways of thinking.

Descriptive vs Critical Thinking

Descriptive	**Critical**
states what happened	identifies the significance
gives the story so far	weighs up one piece of information against another
says how to do something	argues a case according to the evidence
notes the methods used	identifies whether something is appropriate or suitable
explains how something works	indicates why something will work (best)
says when something occurred	identifies why the timing is of importance
lists in any order	structures information in order of importance
states links between items	shows the relevance of links between pieces of information
gives information	draws conclusions

Case Study - United Flight 173

- 1978 United Flight 173
- Problem with landing gear on approach to Portland.
- Went into holding pattern while crew determined what to do – they had an extensive manual, protocols, SOPs and warning gauges, constant checking on passengers, control tower etc
- Were required to yield to other incoming flights extending their time in the air beyond fuel capacity.
- Eight passengers, flight attendant and engineer where killed and twenty-four people with serious injuries.
- Command structure resulting in a breakdown of team work.
- FAA investigation found despite increased internal controls, technological advances, number of dials and lights, safety procedures fatalities increasing – checklist based efficiency.
- Aviation faced a choice: i) increase checklists, tighter internal controls, explicit reminders or ii) focus on risk adaptation by changing the culture within the cockpit.
- Result – Crew Resource Management
- Build flexible systems – cant control and plan for everything.

People:

Heuristics & Risk Perception

Class Discussion

Spent 10mins discussing in a small group how you perceive risks?

1. What influences your perception?

2. How do you know your judgement is accurate?

3. Do you believe you can predict risks?

Report back to class and justify your answers.

Forecasting & Predicting Risk

- How good are you at predicting and assessing risk?

- We are notoriously poor at forecasting events (Slovic, Fischhoff & Lictenstein, 1982; Tetlock, 2006).

Heuristics

- Simple, efficient rules used to generate a judgment or decision, especially for complex problems, (Kahneman et al, 1982).
- People rely on simplified heuristics while judging under uncertainty.
- Provide readily accessible information to guide problem solving and to identify and explore assumptions, questions, or solutions.
- Not a guarantee that the most accurate solution will be determined, rather lead to "best guesses," resulting in quick and easy outcomes based on simple heuristic strategies.

Heuristics - examples

Anchoring - base off the first piece of information we hear / see

Availability – judgement about probability of events based on the ease of recalling other similar events (we place disproportionate weight on what we can easily recall).

Representative (rule of typical things) – mental shortcut to determine probability when faced with uncertainty – judgements based on prototypes we hold. (Tversky & Kahneman, 1974 & 1982

WYSIATI – What You See Is All There Is - overconfidence in making decisions or drawing conclusions from one-sided evidence even when a person knows the evidence to be one sided – facilitates coherence and cognitive ease causing us to accept a statement as true.

Herbet Simon (1978) "rule of thumb"

See the Von Restroff effect – list of items where one written in green.

See Tversky and Kahneman (1982) Linda experiment – important lesson for perception / judgements about risk.

Availability & Risk

Which incident leads to the highest number of deaths in the United States:

- All accidents or the flu?
- car accidents or unintentional falls?
- terrorist attacks or Alzheimer's Disease?

Worldwide:

- Ebola or Malaria?

(Source: *U.S. Centers for Disease Control and Prevention (CDC))*

Representative model

Linda Experiment (Tversky & Kahneman 1982)

- Linda – outspoken, single, very bright. Majored in Philosophy at university and deeply concerned about issues such as discrimination, social justice, attended anti-nuclear demonstrations.

- How likely is it that Linda is a:
  - Teacher
  - Works in a bookstore and takes yoga
  - Active in the feminist movement
  - Is a social worker
  - Member of League of Women voters
  - Bank teller
  - Insurance salesperson
  - Is a bank teller and active in the feminist movement

System 1 vs System 2

Think of our mind as having two systems of thought for assessing, judging and making decisions about the world around us:

System 1: operates automatically, effortlessly and no sense of voluntary control.

System 2: slower, concentration, effortful to a task, complex computations.

(Source: Kahneman, D. (2012). *Thinking, Fast and Slow*. Penguin Books

System 1	System 2
Operates automatically, no sense of voluntary control, effortlessly originating impressions. System 1 can: - Detect one object is more distant than another. - Orient to the source of a sound. - Detect hostility in a voice. - Answer 2 + 2. - Drive a car on an empty road. - Complete the phrase “bread and…” Share these innate skills with animals. We are born prepared to perceive the world around us, recognise objects, orient attention, avoid losses, and fear spiders.	Requires attention, makes choices, completes complex computations, decides what to think about what to do. System 2 can: - Brace for the starter gun in a race. - Focus on the voice of a particular person in a crowded and noisy room. - Look for a woman with white hair. - Search memory to identify a surprising sound. - Monitor the appropriateness of your behavior in a social situation. - Tell someone your phone number. - Fill out a tax form. - Maintain a faster walking speed than is natural for you. - Calculate 17 x 24. In these situations you must pay attention, and you will perform less well, or not at all, if you are not ready or your attention is distracted.

Heuristics

A bat and ball costs $1.10
The bat costs one dollar more than the ball. How much does the ball cost?

The Mind Competition

- A competition between system 1 & system 2 thinking…where system 1 can dominate and either be a force for good or a force for jumping to assumptions.

- How dangerous or advantageous could this be for assessing risks?

WYSIATI

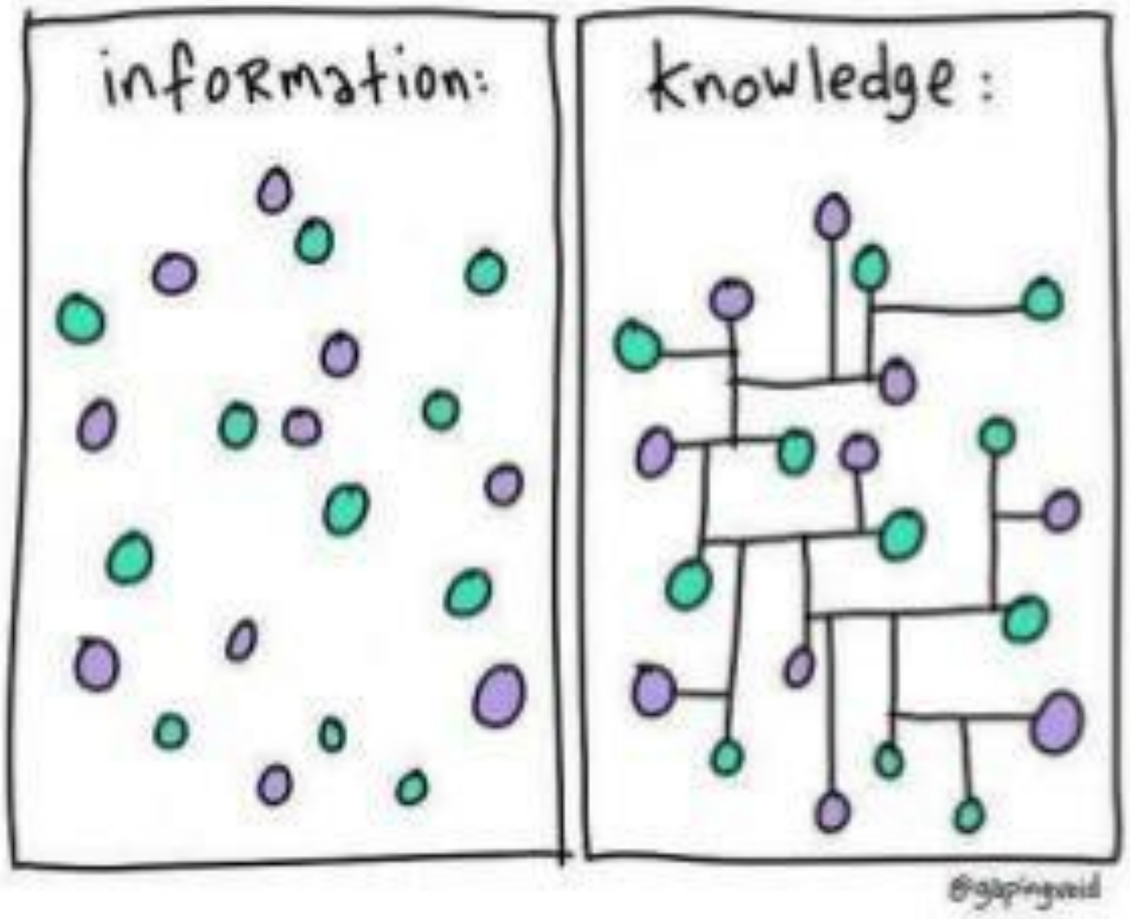
WYSIATI - (What You See Is All There Is)

Will Sally be a good leader? She is intelligent and strong…”

What you see is all there is (WYSIATI)

"We tend to develop stories based on the information at hand ...often without asking, 'what information am I missing'?."

- Daniel Kahneman



e.g. Role model focus often on 'what like' not 'how work'

System 1 vs System 2 Thinking

- Effortless vs effortful thinking.
- One intuitive (recognition) the other rational.
- Why is this concept of how we think important for assessing risk?

System 1 & System 2

Conventional approaches locked into a mindset that assumes:

- If nothing happened, organisation must be doing the right thing.
- Emphasis on day-to-day reporting and operational proficiency, internal checks and balances rather than strategy that appreciates value creation.
- Compliance confined by rigidity where deviation is punished.
- Continual stability through predictability and elimination of risks.
- Enemy thinks like us, tomorrow same as today, pursuit of fixed solutions; and
- Endorses ideas or conclusions because the problem appears so obvious…that is
  adopts a lazy system 2.

  Level of activity is a measure of success – doing things vs doing the right thing.

Are you continually allowing system 1 to dominate?

Complicated versus Complex

It is difficult to avoid the impact of complexity when considering risk—complicated and complex are not the same thing.

The following table represents complicated and complex systems in relation to culture.

Complicated versus Complex

Complicated system	**Complex system**
Comprised of moving parts.	Interaction between nodes or connections within a system.
One part is dependent on the next interaction.	Non-linear.
Generally predicatble.	Emergence of new behaviours or connections.
Mechanistic process (watch, car, machine), akin to Fredrick Winslow Taylor management.	Concerned with behaviours that can emerge over time.
	Butterfly effect or Lorenz system—small changes can lead to big differences or impacts.

Heuristics – its about people

Visions, missions, objectives and strategy – all ultimately managed by humans…not objects, not hand-held devices, not drones…a flashing light is only an indication of a threat or hazard. Our perception can have a critical impact on organisation's culture.

Culture – its about people!

“While technology facilitated many of our advances, the biggest determinant of our ability to communicate and act with speed was in fact cultural—how all the diverse members and units of the task force felt about their participation. We faced a cultural challenge familiar to many businesses: Our force was very good, but this was not enough. They had to believe that it didn’t matter if we were excellent if we weren’t winning.” (GEN (ret.) Stanley McChrystal)

This week

We explored this week, risk is ultimately managed by humans, not objects, and our perception and views of risk can have a serious impact on risk management. This discussion builds on the idea introduced last week that risk is not always negative.

Thinking this through will help to develop your understanding of risk in preparation for Assignment 1: Perceptions of risk in a complex world.